TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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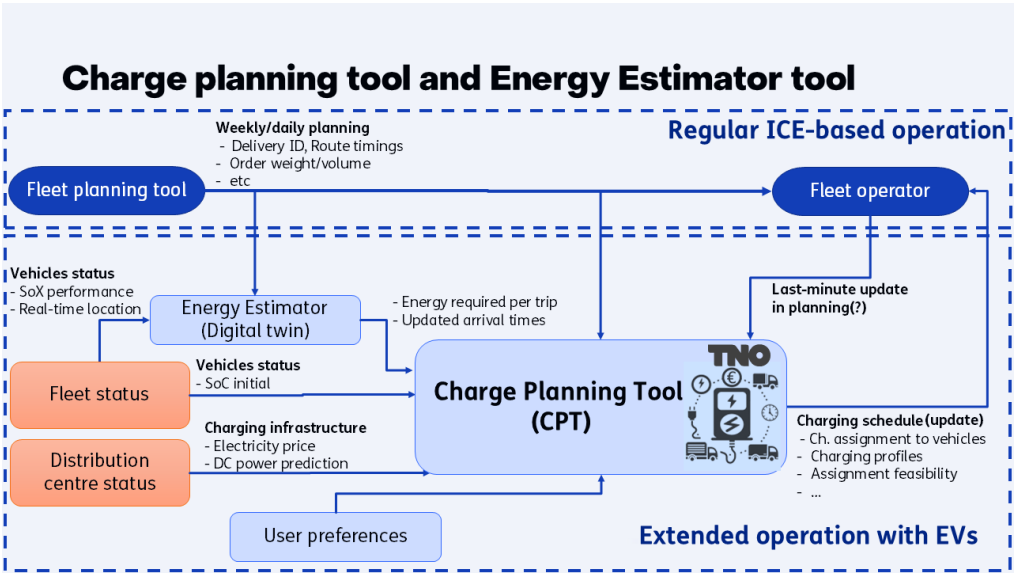
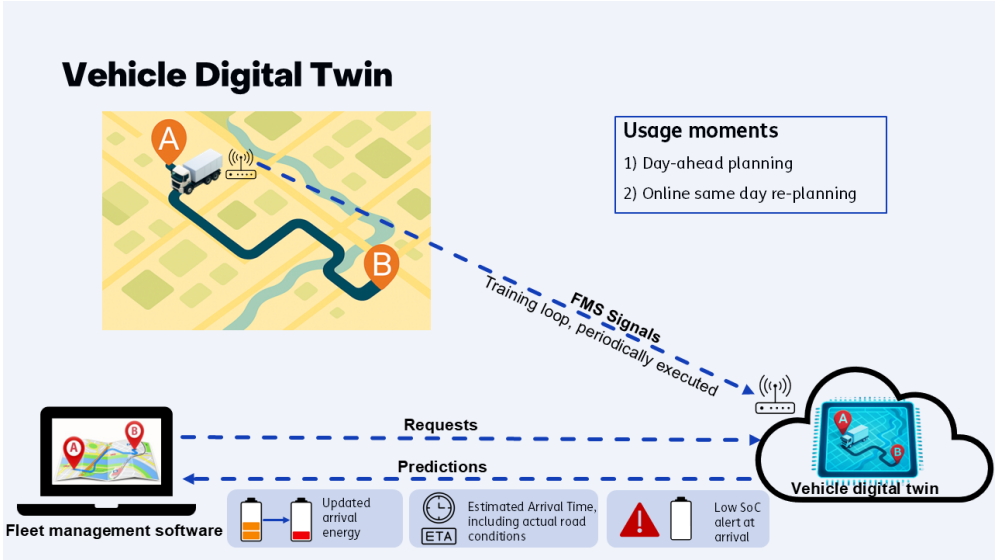
2. Technology Overview

Title of Proposed Technology	Charge Planning Tool and Vehicle Digital Twin for High-Power DC Charging Fleets			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-generation Energy (), Bio-health ( ), Aerospace ( ), Quantum ( ), Advanced Robotics & Manufacturing ( ), AI (√), Cybersecurity (), Hydrogen ( ), Semiconductor materials ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communications		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Internet SW	Batteries	Secondary Batteries
Concept of Proposed Technology	This proposal develops an integrated software platform that enables (i) a smart Charge Planning Tool (CPT) and (ii) a vehicle-level Digital Twin (DT) to optimise charging operations for fleets using ultra-fast DC chargers with power ratings of 240 kW or higher.			

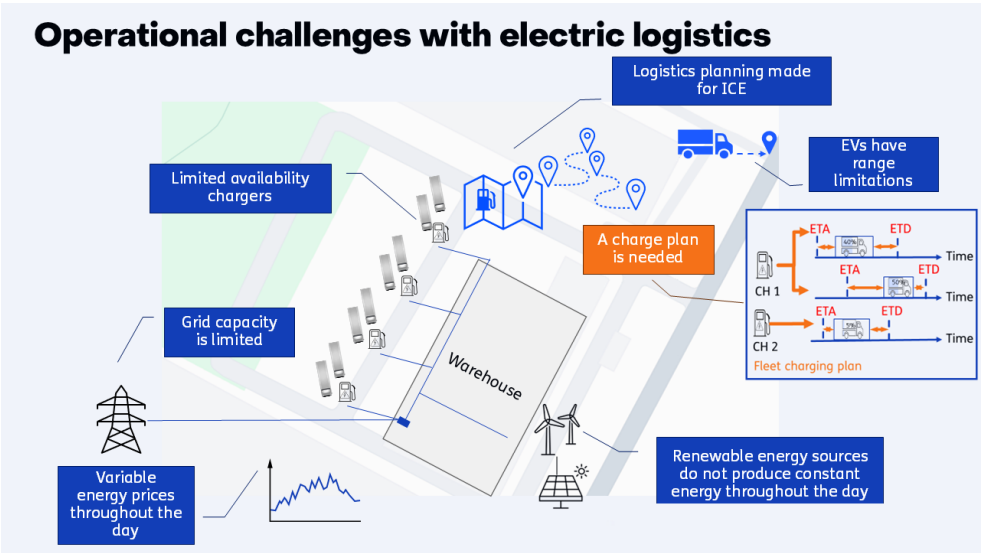


The DT predicts vehicle energy use, arrival times, and charging time and power profiles under real-world conditions, including temperature and battery ageing effects. The CPT uses these DT predictions to schedule charging sessions (time, location, power) under constraints such as limited grid connection capacity, charger availability, and vehicle duty cycles. [R1][R2][R3][R4]



The platform is designed to interface with charger management systems using open protocols such as OCPP, and supports vehicle-side smart charging workflows based on ISO 15118 where available. [R10][R11]

The Korean SME partner as a charging systems manufacture, will contribute charger domain knowledge, integration support and pilot demonstrations.

Rationale for Joint R&D Project	 Electrification of heavy-duty and commercial fleets adds a new planning dimension: charging must be scheduled alongside routes and delivery constraints, while respecting grid capacity and charger availability. TNO has demonstrated Charge Planning Tool concepts for fleet operations and is developing digital twins and planning tools in the Charging Energy Hubs context. [R1][R2][R3][R4] High-power DC charging can accelerate battery ageing if power profiles and dwell times are not managed. Incorporating a battery-aware digital twin in the planning loop enables schedules that meet operational constraints while reducing degradation and charging cost. [R5][R6][R7] By combining the Korean SME's charger expertise and pilot access with TNO's modelling, optimisation and validation facilities, the project targets a TRL-6 demonstrator of degradation-aware charge planning and digital twinning that is directly relevant to European fleet operators and charging infrastructure providers.
Key Words	1. Charge planning tool 2. Vehicle digital twin 3. Smart charging 4. High-power DC fast charging 1. Battery ageing and degradation-aware optimisation

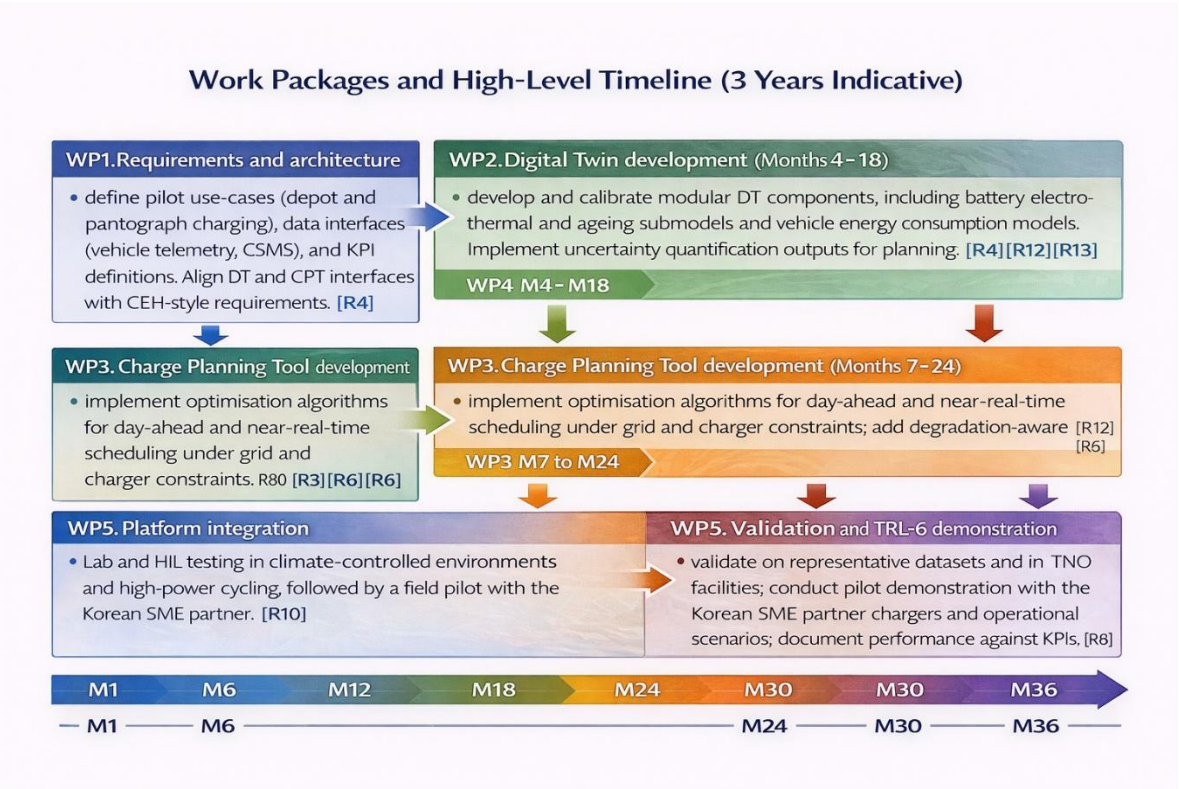
3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	1) Develop a vehicle Digital Twin (DT) that predicts energy consumption, SoC at arrival, and charge-time and power profiles under varying ambient and battery ageing conditions. [R4][R12][R13] 2) Develop a Charge Planning Tool (CPT) that optimises charging schedules for a non-homogeneous fleet with multiple charging opportunities, accounting for charger availability, charger-vehicle compatibility, and grid constraints. [R3][R4] 3) Extend the optimisation with degradation-aware objectives using battery ageing models, aiming to reduce calendar and cyclic ageing drivers during charging operations. [R5][R6][R7] 4) Integrate CPT with charger management systems via OCPP and support ISO 15118 based smart charging interactions where feasible in pilots. [R10][R11] 5) Validate the integrated DT and CPT using TNO facilities and representative fleet and charging data, reaching TRL-6 demonstration. [R8][R9]			
R&D Target Specifications	DT prediction performance (fleet planning inputs): • Predict charging energy demand with at least 5% better accuracy than a rule-based baseline, as targeted in CEH digital twin requirements. [R4] • Provide SoC at arrival and ETA predictions with uncertainty indication (95% confidence bound) suitable for planning optimisation. [R4] • Provide charge-time and remaining power profile estimates at least every 15 minutes during charging sessions in hub contexts. [R4] CPT optimisation capability: • Generate feasible charging schedules for fleets with fewer chargers than vehicles and multiple charging opportunities. [R3] • Respect grid connection limits and charger-vehicle compatibility constraints. [R3][R4] • Support objectives for charging cost minimisation and battery degradation mitigation based on ageing proxies. [R5][R6] Integration and interoperability: • Interface with Charge Station Management Systems using OCPP messaging for scheduling and monitoring. [R10] • Where supported by the pilot environment, enable ISO 15118 based smart charging signals for vehicle side constraints. [R11]			

Validation environment:

- Demonstrate the integrated DT and CPT using climate-controlled battery and vehicle-level testing capabilities and a pilot charging site (depot and or pantograph) provided by the Korean SME partner. [R8][R9]

R&D Plan



Work packages and high-level timeline (3 years indicative):

WP1 Requirements and architecture (Months 1-6): define pilot use-cases (depot and pantograph charging), data interfaces (vehicle telemetry, CSMS), and KPI definitions. Align DT and CPT interfaces with CEH-style requirements. [R4]

WP2 Digital Twin development (Months 4-18): develop and calibrate modular DT components, including battery electro-thermal and ageing submodels and vehicle energy consumption models. Implement uncertainty quantification outputs for planning. [R4][R12][R13]

WP3 Charge Planning Tool development (Months 7-24): implement optimisation algorithms for day-ahead and near-real-time scheduling under grid and charger constraints; add degradation-aware objective terms. [R3][R5][R6]

WP4 Platform integration (Months 16-30): implement OCPP-based integration with CSMS and a scheduling API; integrate DT predictions into CPT; implement dashboards for operators. [R10]

WP5 Validation and TRL-6 demonstration (Months 24-36): validate on representative datasets and in TNO facilities; conduct pilot demonstration with the Korean SME partner chargers and

	operational scenarios; document performance against KPIs. [R8][R9]

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**TNO (Research Institute) expected roles and responsibilities:** • Project leadership and system architecture for the integrated Charge Planning Tool (CPT) and vehicle Digital Twin (DT). • Development and calibration of physics-based and hybrid (physics plus data) DT models for energy use, SoC and charge-time prediction, including uncertainty quantification. [R4][R12][R13] • Development of optimisation algorithms for charging schedules under grid and charger constraints, extending toward degradation-aware charging objectives. [R3][R5][R6] • Integration design for interoperability with Charge Station Management Systems via OCPP and smart charging workflows via ISO 15118 when available. [R10][R11] • - Validation and demonstration using TNO Battery Lab and test facilities (climate-controlled testing up to vehicle level) and performance evaluation against defined KPIs. [R8][R9]
Expected R&R of Korean SME Partner	**Expected roles and responsibilities of Korean SME partner:** • Provide expertise and access related to high-power DC charging systems, including depot and pantograph charging solutions. • Provide charger-side interfaces and integration support for linking the CPT to a Charge Station Management System (CSMS), including OCPP configuration in pilot settings. [R10] • Provide representative charging session logs, charger telemetry and operational constraints from field deployments for model training and validation (subject to agreed data governance). • Host and support pilot demonstrations (hardware availability, installation and commissioning support, safety procedures) to reach TRL-6 demonstration conditions. • - Contribute to commercialisation planning, including product integration pathway into the SME's charger and infrastructure offerings.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected outcomes:** • Prototype integrated Charge Planning Tool (CPT) and vehicle Digital Twin (DT) software stack suitable for integration with fleet planning and charger management systems. [R2][R3][R4] • Validated DT prediction services for ETA, SoC at arrival, charging energy demand and charge-time estimation with uncertainty outputs for operational planning. [R4] • Degradation-aware charging optimisation capability demonstrated for high-power DC charging use-cases, informed by published charging and degradation studies. [R5][R6][R7] • Interoperability demonstrator using OCPP based communication between the CPT and CSMS, and optional ISO 15118 based smart charging signals where supported. [R10][R11] • TRL-6 field-relevant demonstration using the Korean SME partner charging hardware (including pantograph and depot chargers) and representative fleet scenarios, supported by TNO validation facilities. [R8][R9] Expected impact: • Improved utilisation of constrained grid connections and charging infrastructure through coordinated scheduling, supporting zero-emission fleet operations. [R1][R2] • Reduced operational cost and reduced battery wear risk via degradation-aware charging decisions, improving total cost of ownership for fleet operators. [R5][R6] • - Strengthened market position for the Korean SME through an advanced software layer for smart charging and digital twinning compatible with European fleet needs and protocols. [R1][R10]

6. References

[R1] TNO (2024-12-05). "Optimal charging planning for logistics" (Charge Planning Tool insight). Accessed 2025-12-19. https://www.tno.nl/en/newsroom/insights/2024/12/optimal-charging-planning-logistics/

[R2] TNO. "Charging Energy Hubs" (project page describing digital twins and charging planning tools). Accessed 2025-12-19. https://www.tno.nl/en/sustainable/mobility-logistics/charging-energy-hubs/

[R3] Beckers, A.; Medina, R.; Wilkins, S. (2024). "Charge Planning Tool for Heavy-Duty Electric Vehicle Fleets" (conference paper record). TNO Publications Repository. Accessed 2025-12-19. https://resolver.tno.nl/uuid:4043c8dc-e109-4441-8b23-0ea3a2fddf68

[R4] Beckers, A. (2025). "Requirements and Specification of the Operational Fleet Management Support Tools" (Charging Energy Hubs, D2.4). TNO Publications. Accessed 2025-12-19. https://publications.tno.nl/publication/34644650/KVx1RrIc/beckers-2025-D2.4charging.pdf

[R5] Rath, S.; Medina, R.; Wilkins, S. (2023). "Real-time Optimal Charging Strategy for a Fleet of Electric Vehicles Minimizing Battery Degradation". TNO Publications. Accessed 2025-12-19. https://publications.tno.nl/publication/34641555/EemswK/rath-2023-real.pdf

[R6] Geerts, D.C.; Medina, R.; Wilkins, S.; et al. (2022). "Optimal charging of electric vehicle fleets: Minimizing battery degradation". TNO Publications. Accessed 2025-12-19. https://publications.tno.nl/publication/34640417/k3zIoq/geerts-2022-optimal.pdf

[R7] Rath, S.; Beckers, A.; Netto, P.; Medina, R.; Wilkins, S. (2025-06). "Charge Scheduling for a Fleet of Electric Vehicles using Battery Digital Twin" (EVS38 proceedings). Accessed 2025-12-19. https://evs38-program.org/images/Proceedings/D%20Charging%20Infrastructure%20and%20grid%20integration/197_Charge%20Scheduling%20for%20a%20Fleet%20of%20Electric%20Vehicles%20using%20Battery%20Digital%20Twin.pdf

[R8] TNO. "Batteries" (includes Battery Lab facilities overview). Accessed 2025-12-19. https://www.tno.nl/en/sustainable/mobility-logistics/batteries/

[R9] Van de Putte, J.; et al. (2022). "TNO, connecting people and knowledge" (Battery lab test capabilities). TNO Publications. Accessed 2025-12-19. https://publications.tno.nl/publication/34639492/gxMYkR/putte-2022-vehicle.pdf

[R10] Open Charge Alliance. "Open Charge Point Protocol (OCPP)" (protocol overview). Accessed 2025-12-19. https://openchargealliance.org/protocols/open-charge-point-protocol/

[R11] CharIN. "ISO/IEC 15118 Communication Standard" (standard overview for smart charging and Plug and Charge). Accessed 2025-12-19. https://www.charin.global/technology/iso15118

[R12] Eindhoven University of Technology (TU/e). "Steven Wilkins" (research profile). Accessed 2025-12-19. https://research.tue.nl/en/persons/steven-wilkins/

[R13] Wilkins, S. (2013). "Enhanced battery model including temperature effects" (EVS27 record). TNO Publications Repository. Accessed 2025-12-19. https://repository.tno.nl/SingleDoc?find=UID%2039157

[R14] Polat, H.; Hosseinabadi, F.; Hasan, M.M.; et al. (2023-02-08). "A Review of DC Fast Chargers with BESS for

Electric Vehicles" (includes grid and charging infrastructure context; co-author Steven Wilkins). TNO Publications mirror of the open access paper. Accessed 2025-12-19. https://publications.tno.nl/publication/34640889/so7Rus/polat-2023-review.pdf

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization